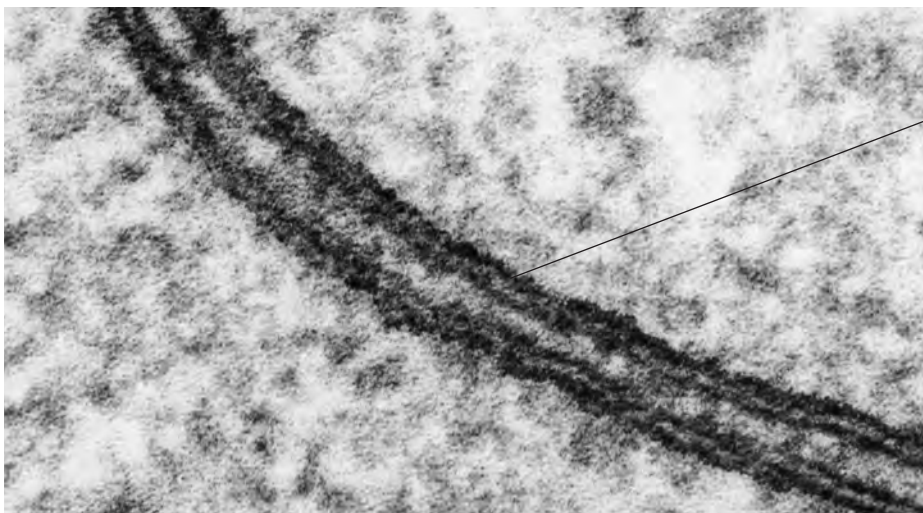


2 All cells are surrounded by a membrane.

The photograph shows two animal cells where their membranes meet.



(Source: © Don W. Fawcett / Science Photo Library)

(a) Name **three** types of molecule present in structure **A**.

(2)

- 1
- 2
- 3

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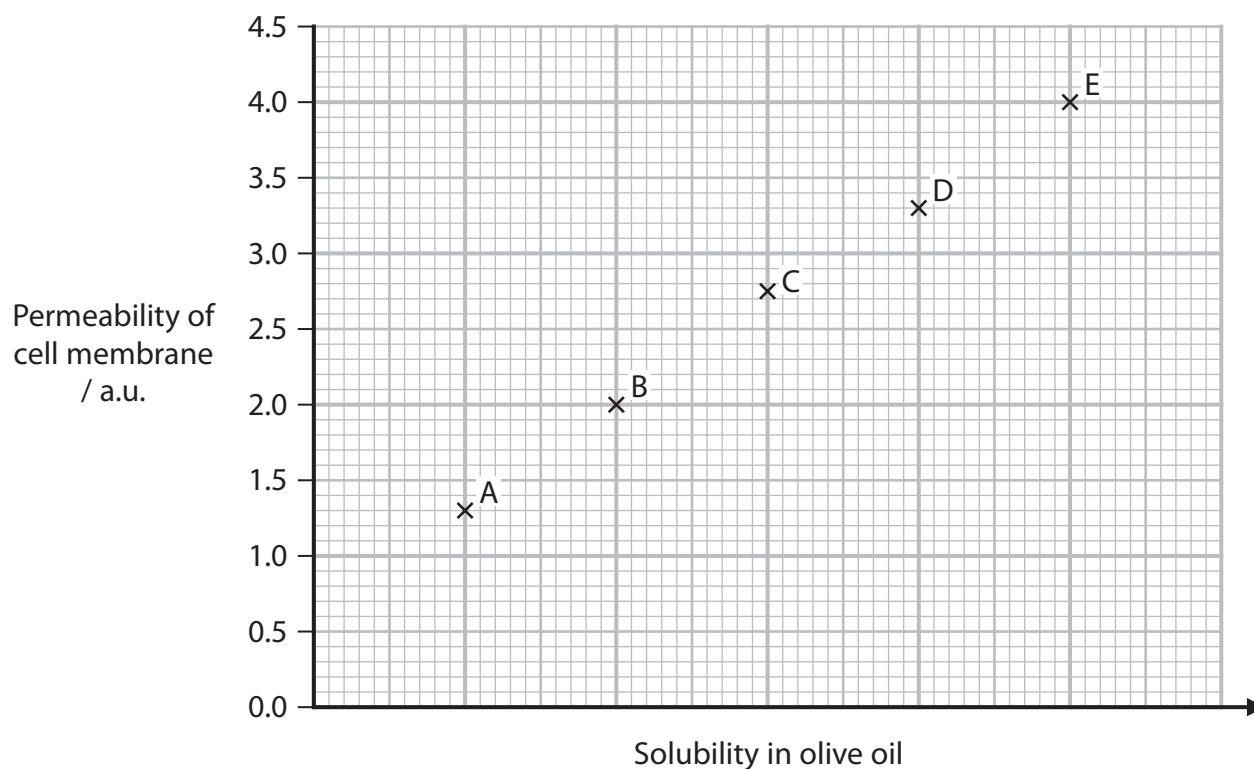


- (b) Some of the evidence for the structure of cell membranes comes from studies of permeability.

The solubility of substances A to E in olive oil was determined.

The permeability of a cell membrane to these five substances was also measured.

The graph shows the results.



Explain the results shown in this graph.

(3)

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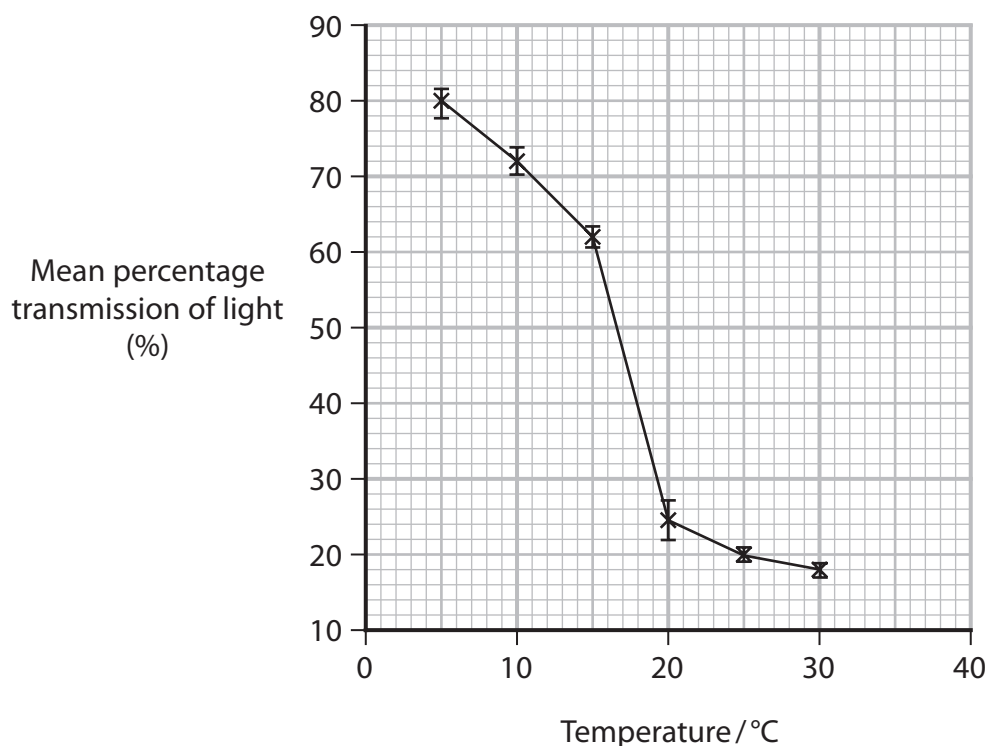
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- (c) Beetroot cells contain a water-soluble pigment. The effect of increasing temperature on the release of this pigment was investigated.

The quantity of pigment released was measured using a colorimeter, an instrument that can measure the transmission of light through a coloured solution.

The transmission of light through the solution decreases as the intensity of the colour increases.

The results are shown in the graph.



- (i) Explain the results of this investigation.

(3)

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- (ii) Describe a procedure that could have been used to obtain the results shown in this graph.

(5)

(Total for Question 2 = 13 marks)

